or **Link reaction**. The reaction of pyruvate oxidation is

$$2x\ CH_3COCOOH + 2NAD^+ + 2CoA \xrightarrow[\text{complex/ } Mg^{++}]{\text{Pyruvate dehydrogenase}} 2xCH_3CO.CoA + 2NADH + 2H^+ + 2CO_2\uparrow$$

Pyruvate dehydrogenase complex consist of three distinct enzymes, such as

1. Pyruvate dehydrogenase
2. Dihydrolipoyil transacetylase
3. Dihydrolipoyil dehydrogenase

and five different coenzymes, TPP (Thymine Pyro Phosphate), $NAD^+$, FAD, CoA and lipoate.

### 14.5.3 Krebs cycle or Citric acid cycle or TCA cycle:

Two molecules of acetyl CoA formed from link reaction now enter into Krebs cycle. It is named after its discoverer, German Biochemist **Sir Hans Adolf Krebs** (1937). The enzymes necessary for TCA cycle are found in mitochondrial matrix except succinate dehydrogenase enzyme which is found in mitochondrial inner membrane (Figure 14.7).

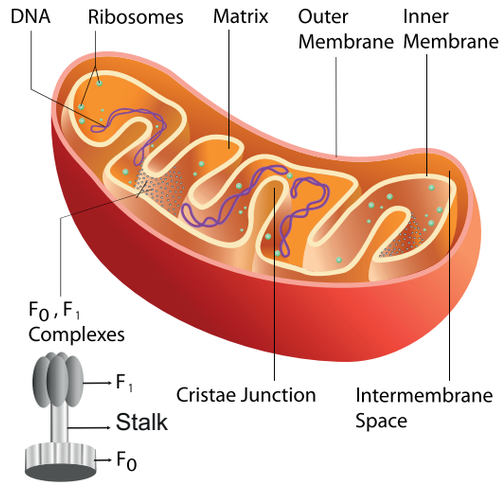
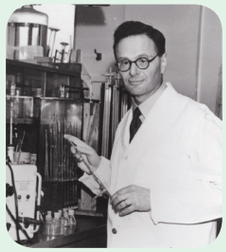



**Figure 14.7:** Structure of Mitochondrion



**Sir Hans Adolf Krebs** was born in Germany on 25th August 1900. He was awarded Nobel Prize for his discovery of Citric acid cycle in Physiology in 1953.

TCA cycle starts with condensation of acetyl CoA with oxaloacetate in the presence of water to yield citrate or citric

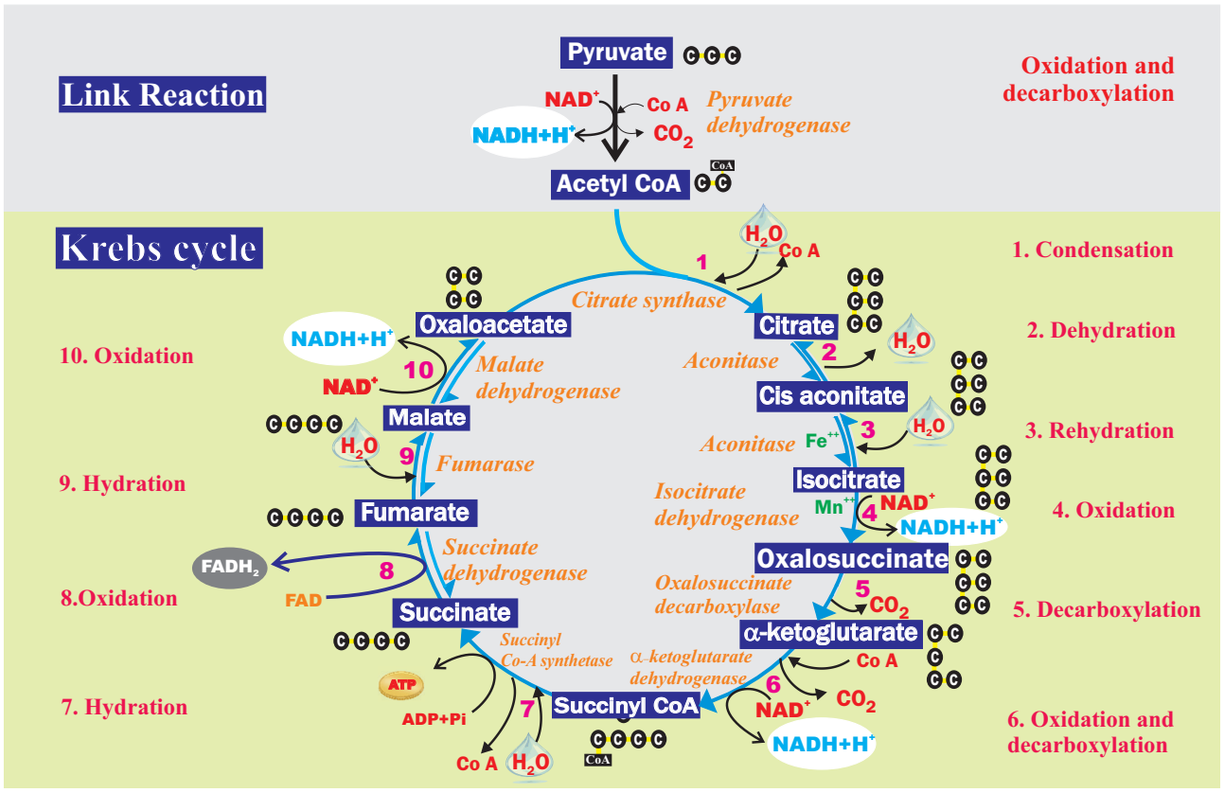



**Figure 14.8:** Krebs cycle or Citric acid cycle



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